

Problem Set 4 (45 pts)

Due at 11:59pm

Friday, September 26

The first three problems are from <https://aimacode.github.io/aima-exercises/> (possibly) with minor typo fixes, slight rewriting, or addition of missing information.

1.(15 pts) (Exercise 4.8) This exercise explores subset–superset relations between belief states in sensorless or partially observable environments.

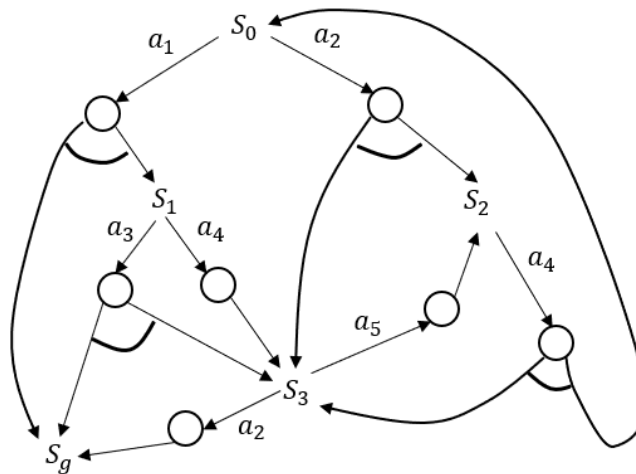
(a) (2 pts) Argue that if an action sequence is a solution for a belief state b , it is also a solution for any subset of b .

(b) (5 pts) Can anything be said about supersets of b in (a)?

(c) (4 pts) Explain in detail how to modify graph search for sensorless problems to take advantage of your answers in (a) and (b).

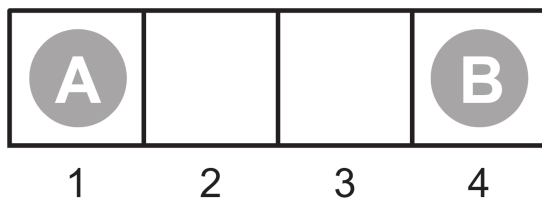
(d) (4 pts) Explain in detail how to modify and–or search for partially observable problems, beyond the modifications you describe in (c).

2.(8 pts) Shown below is the AND-OR graph representation of the state space of a search problem, which has five states S_0, S_1, S_2, S_3, S_g and five actions a_1, a_2, a_3, a_4, a_5 . Here, S_0 is the initial state and S_g is the goal state. (The structure is a graph because a state can be reached from multiple other states.) It is known that some of the actions may be erratic in non-deterministically generating an outcome state.



Give a solution to this problem. You may draw a closed contour around the subgraph that represents the solution. Or you may write down a conditional plan that uses atomic statements which are actions, and **if-then** or **if-then-else** statements whose conditions are states.

- 3.(22 pts)** (Exercise 5.8) Consider the two-player game with its starting position shown in the figure below. Player A moves first. The two players take turns moving, and each player must move his token to an open adjacent space in either direction. If the opponent occupies an adjacent space, then a player may jump over the opponent to the next open space if any. (For example, if A is on 3 and B is on 2, then A may move back to 1.) The game ends when one player reaches the opposite end of the board. If player A reaches space 4 first, then the value of the game to A is $+1$; if player B reaches space 1 first, then the value of the game to A is -1 .



- (a) **(7 pts)** Draw the complete game tree, using the following conventions:
- Write each state as (s_A, s_B) , where s_A and s_B denote the token locations.
 - Put each terminal state in a square box and write its game value in a circle.
 - Put *loop states* (states that already appear on the path to the root) in double square boxes. Since their value is unclear, annotate each with a “?” in a circle.
- (b) **(3 pts)** Now mark each node with its backed-up minimax value (also in a circle). Explain how you handled the “?” values and why.
- (c) **(6 pts)** Explain why the standard minimax algorithm would fail on this game tree and briefly sketch how you might fix it, drawing on your answer to (b). Does your modified algorithm give optimal decisions for all games with loops?
- (d) **(6 pts)** This 4-square game can be generalized to n squares for any $n > 2$. Prove that A wins if n is even and loses if n is odd.